



Enhancing multivariate time-series anomaly detection with positional encoding mechanisms in transformers

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Abstract

The surge in automation driven by IoT devices has generated extensive time-series data with highly variable features, posing challenges in anomaly detection. DL, particularly Transformer networks, has shown promise in addressing these issues. However, Transformer networks struggle with accurately determining the position of data points and maintaining the order of data in sequences, leading to the development of Positional Encoding (PE). Initially, Absolute PE was introduced, but newer methods like Relative PE and Rotary PE have been adopted in natural language processing tasks to improve performance. This study evaluates the potential of PEs including Absolute PE, Rotary PE, and two modifications of Relative PE methods (Representative attention and Global attention), for multivariate time-series anomaly detection problems. The experimental results indicate that Absolute PE, with a 98% accuracy score, performs well across different window sizes. Representative attention, with a 98% F1-score, performs best for short sequences (8, 16, and 32); whereas, Global attention, with a 97% F1-score, is more effective for longer sequences (64 and 128). Additionally, Absolute PE has the shortest training times, starting at 25 for sequence length 8 and increasing to 192 for length 128. Rotary PE also has slightly longer training times compared to Absolute PE. On the other hand, Representative attention consistently has the longest times, starting at 48 for length 8 and reaching 366 for length 128. Overall, Absolute PE and Global attention are the most time-efficient; while, Representative attention has significantly higher training times, particularly for long sequences.

Keywords Positional encoding · Transformer · Internet of things · Multivariate time-series · Anomaly detection

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1 Introduction

The term “Internet of Things” (IoT) signifies a major advancement in industrialization and urbanization, reshaping numerous sectors by connecting devices, objects, and systems capable of sharing data autonomously [1]. Driven by internet expansion, hardware miniaturization, data analytics advancements, and cloud computing, IoT has become feasible and cost-effective, finding applications in smart homes, transportation, health systems, energy consumption, and security [2, 3]. IoT devices generate vast amounts of data, known as big data, categorized as structured, semi-structured, and unstructured [4]. Big data analytics, using modern algorithms, helps organizations extract useful patterns and gain competitive advantages. Analyzing sequential data, which is a complex form of big data, is crucial for practical applications.

Sequential data consists of a list of elements with corresponding time stamps, forming an ordered collection of samples $S = \{S_1, S_2, S_3, \dots, S_n\}$, where each item $S_i = \{X_i^1, X_i^2, X_i^3, \dots, X_i^D\}$ represents a single instance with D features, which can include multiple values per time [5]. Real-valued time-series, such as stock prices or clinical vital signs, are crucial forms of sequential data, defined as $\{x_1, x_2, x_3, \dots, x_T\}$, with observations taken over time intervals (e.g., monthly, daily) [6].

Time-series data can be univariate or multivariate from an observational point of view. Sensors often generate multiple values simultaneously, necessitating the analysis of these values’ relationships to achieve desired goals. This can be done effectively by considering individual series as components of multivariate time-series, denoted as Z_t , a $k \times 1$ vector at time t , and analyzing them jointly [7]. Multivariate time-series analysis helps understand dynamic relationships and improves accuracy by utilizing information from related series. One of the common problems in time-series analysis is anomaly detection, which is being particularly important for providing accurate and timely insights to decision-makers [8]. Anomaly detection identifies data points deviating from the norm, which is crucial for distinguishing between normal and abnormal data. In time-series data, anomaly detection involves establishing normal behavior, detecting deviations, and triggering alerts [9]. Adaptation and feedback mechanisms maintain algorithm accuracy in dynamic environments. Despite challenges, real-time anomaly detection is vital in finance, healthcare, manufacturing, and cybersecurity for risk mitigation, operational optimization, and informed decision-making [10].

Anomaly detection in time-series data is a challenging and active research area with a long history of real-world applications [11]. According to a recent survey on anomaly detection for IoT-based time-series data [12], three common methods are used: statistical approaches [11, 13, 14], Machine Learning (ML) methods [15–17], and Deep Learning (DL) algorithms [18–21]. Initially, statistical methods were useful for low volumes of data, but with the increase in data from IoT devices, their effectiveness has diminished. ML algorithms, leveraging basic mathematical concepts, have improved anomaly detection efficiency, but the limitations of classical ML in big data analysis have led to the rise of DL. DL algorithms like Convolutional Neural Network (CNN) [22], Recurrent Neural Network (RNN) [23],

Long Short-Term Memory (LSTM) [18], Gated Recurrent Unit (GRU) [19], Autoencoders [21], inspired by neural networks, are now outperforming other techniques. However, they may struggle with discovering relationships in multivariate time-series and long sequences.

The new advancement in DL is Transformers, which uses attention mechanisms to focus on important data elements. Introduced by Google researchers in 2017 for Natural Language Processing (NLP) tasks [24], Transformers have shown excellent performance in sequential data analysis, including machine translation [25], computer vision [26], and speech recognition [27]. Since their introduction to time-series modeling, Transformers have become the state-of-the-art approach for anomaly detection problems [28]. However, Transformers have a limitation in determining the actual location and order of each observation in a sequence. To address this, Positional Encodings (PEs) are used [24]. Absolute PE, introduced in the “Attention is All You Need” paper, and later enhancements like Relative PE [28, 29] and Rotary PE [30] are considered best for NLP tasks. Relative PE can be directly applied to the query and key vectors in the multi-head attention module, which has led to many modifications for both short and long sequences.

Despite these advancements, it remains unclear which PE method is best for time-series data. This study aims to apply Absolute PE, two commonly used modifications of Relative PE (Representative attention and Global attention), and Rotary PE to the Multivariate Time-series Anomaly Detection (MVTAD) problem and evaluate their performance on both short and long sequences. To our knowledge, there are no proposed Rotary PE and Representative attention mechanisms specifically for MVTAD problems. Furthermore, there are no available comparative studies among the different PEs to assess their potential in the context of MVTAD problems.

Motivated to contribute to the literature, we aim to achieve the following objectives:

- Relative PE can be applied directly to keys and values in attention modules. Therefore, many variations have been introduced in NLP tasks, including Representative attention and Global attention. In this study, for the first time, we apply each of these variations for MVTAD problems and compare their results with Absolute PE and Rotary PE.
- Evaluate the potential of the PEs on sequences of lengths 8, 16, 32, 64, and 128, considering both short and long sequences.
- Utilize a dynamic tuning technique, Optuna, for hyperparameter tuning.

The rest of the paper is organized as follows. Section 2 presents the related works in the literature. Section 3 provides information about the backgrounds of algorithms. Section 4 explains the proposed methods and an illustrative scenario in detail. Experimental analyses are conducted on the illustrative scenario in Sect. 5. Finally, Sect. 6 concludes the paper and recommends future work.

2 Related work

Since adopting Transformers for time-series data, various variations of Transformers are employed. This area of research remains open, and it represents the current state-of-the-art techniques for anomaly detection in both short and long sequences. Due to their attention mechanism, Transformers are highly effective in learning dependencies among features in multivariate time-series data, making them the preferred choice for decision-makers.

One of the initial Transformer-based approaches for anomaly detection was conducted in [31], where the authors proposed reconstruction techniques, incorporating two novel components: the attention mechanism used for updating timesteps in parallel and the masking strategy used to detect anomalies in advanced time. TransAnomaly in [32] is the first study to combine the VAE and Transformer. In [33], Anomaly Transformer utilized Transformer architecture for unsupervised time-series anomaly detection. It incorporates the anomaly attention mechanism to model prior association and series-association, aiming to compute the association discrepancy. In [42], the relative PE (Global attention) is proposed as an improved PE for the attention mechanism with VAE for anomaly detection. Additionally, a graph structure was proposed with a Transformer architecture in [34]. TranAD in [35], utilizes Transformer neural networks with self-conditioning and adversarial training. This mechanism enhances stability with large input sequences and improves the detection of small deviation anomalies. Coupled with model-agnostic meta-learning, TranAD achieves superior performance, outperforming simple Transformer models by over 11 in validation with limited data. The CAT is proposed in [36] for content-aware anomaly detection. In [37], the authors proposed the root square sparse self-attention to decrease the computational complexity of attention and to prevent failing concept drift adaptation method. DCT-GAN in [38] is proposed by utilizing the encoder layer of the Transformer and GAN with a multi-generator and single discriminator. MEMTO is in [39] proposed a bi-dimensional deviation-based detection criterion in online detection, aggregating comprehensive information from both latent and input space. In [40], an unsupervised time-series anomaly detection method using Transformer is proposed, outperforming LSTM and CNN by leveraging self-attention for learning dynamic patterns. The model comprises an encoder with multiple Transformer layers and a decoder with a 1D convolution layer, enabling it to capture multi-level, rich information. In [41], AnoFormer is a pure Transformer-based GAN framework that includes a generator and discriminator is proposed. Also, preprocessing and embedding methods are tailored for effective handling of time-series data. A similar study was conducted by N. Foumani et al. [43], where they evaluated two commonly used PE techniques, including Absolute PE and Relative PE (Representative attention and Global attention), on multivariate time-series data. Additionally, the researchers introduced modifications of both PEs, namely tAPE, and eRPE, to improve time complexity and performance. However, there are three key distinctions between their study and ours. First, they evaluated the proposed PE models on a classification problem; while, we applied them to anomaly detection. Second, they did not evaluate the performance of Rotary PE or compare the results to

determine which PEs perform best. Lastly, specifying the sequence length (or window size) in time-series data is crucial. In their study, the researchers did not specify or experiment with the efficacy of the PEs on both short and long sequences, as we did in our research.

The summary of the detailed literature review conducted on MVTAD, presented in Table 1, identifies and carefully analyzes several related methods. Specifically, we examine their proposed methods, model structures, and the preferred PEs, which are important components in modeling Transformer architectures. We find that some studies employ the Encoder–Decoder Transformer, while others propose only the structure of the Encoder layer, which includes an attention module, feedforward network with residual connections, and normalization process. Additionally, we categorize these methods into four domains: Absolute PE, Relative PE (Representative attention), Relative PE (Global attention), and Rotary PE. One of our main intentions is to determine which of the three well-known PEs is most preferred by researchers and employed in MVTAD. Table 1 clearly shows in the categorized domain most studies utilized Absolute PE, with a limited number utilizing Global attention; while, none proposed Rotary PE and Representative attention. Likewise, the summary of Table 1 encouraged us to adopt PE methods to MVTAD problems, which have been particularly successful in NLP and computer vision in the literature. A comparative analysis was conducted among the best PE methods. Although some of these methods have been tested on time-series data in the literature, to the best of our knowledge, a study like ours has not yet been conducted.

3 Background information

Since being introduced by Google researchers, Transformers have revolutionized artificial intelligence by using the attention mechanism, which calculates self-attention in parallel to remember important parts of the input sequence [24]. The concept of self-attention, shown as pink in Fig. 1, is part of the multi-head attention mechanism that allows a sequence to attend to other elements of the same sequence, capturing dependencies between observations. In self-attention, each data point is associated with three vectors: Query, Key, and Value. These are used to compute a weighted sum of the values based on the similarity between the query and key vectors. This output is then multiplied by the value vectors. Self-attention helps capture long-range dependencies and is particularly useful in anomaly detection. Multi-head attention extends self-attention by applying it multiple times in parallel with different Query, Key, and Value vectors, then concatenating and transforming the outputs to produce the final result. This allows the model to focus on other parts of the input sequence simultaneously, enhancing its ability to capture diverse patterns and relationships within the data.

Transformers map a query and a set of key-value pairs to an output. In other words, for an input series $x_t = \{x_1, x_2, \dots, x_L\}$, self-attention computes an output series $z_t = \{z_1, z_2, \dots, z_L\}$ where $z_i \in \mathbb{R}^d$ and is computed as a weighted sum of input elements through Eqs. 1–3:

Table 1 Summary of existing transformer-based studies indicating the different PEs for IoT-based MVTAD

Domain	Reference	Year	Methods	Model structure	Dataset
Absolute PE	Meng et al. [31]	2019	Reconstruction-based model defines a threshold to identify the difference between the input and the reconstructed data	Encoder with masking	NASA spacecraft
	Zhang et al. [32]	2021	The encoder is used for prediction purposes; while, the VAE is utilized to verify the difference between the predicted ad input data	Encoder with VAE	SMAP, MSL and SMD
	Xu et al. [33]	2021	Incorporating the anomaly-attention mechanism, it models prior association and series-association to compute the Association Discrepancy for unsupervised time-series anomaly detection	Encoder	SMD, PSM, MSL, SMAP, ISA
	Chen et al. [34]	2022	It includes automatic graph structure learning, graph convolution, and temporal dependency modeling using a Transformer-based architecture. The connection learning policy uses Gumbel-softmax sampling for bidirected sensor links. Influence propagation convolution describes anomaly information flow between nodes. To address quadratic complexity, a multi-branch attention mechanism replaces the original multi-head self-attention method	Encoder	SWaT, WADI, SMAP, MSL and NASA datasets
	Tuli et al. [35]	2022	A prediction-based approach, TranAD, uses Transformer neural networks with self-conditioning and adversarial training. This enhances stability with large input sequences and improves detection of small deviation anomalies. Coupled with model-agnostic meta-learning, TranAD achieves superior performance	Encoder	NAB, UCR, MBA, SMAP, MSL, SWaT, WADI, SMD, MSDS
	Zhang et al. [36]	2022	In CAT, a prediction-based approach, the encoder learns event sequence representations with content awareness, and the decoder embeds sequences for anomaly detection. The content awareness layer processes event content to generate representations. The encoder produces feature maps; while, the decoder includes a token to indicate sequence status	Encoder	HDFS, BGL and Thunderbird
	Ding et al. [37]	2022	It introduces an online sparse Transformer blending online learning with the Transformer. It learns from past data and adapts to new concepts through fine-tuning. Presented a concept drift adaptation method for changes in data distribution and anomaly patterns. Proposed root square sparse self-attention to reduce time and memory complexity to $O(L\sqrt{L})$	Encoder with an extra online learning layer	Yahoo and NAB

Table 1 (continued)

Domain	Reference	Year	Methods	Model structure	Dataset
	Li et al. [38]	2022	The DCT-GAN architecture improves accuracy and robustness in the existing SOTA model for the anomaly detection problem. It uses multiple Transformer-based generators of various scales to capture both coarse- and fine-grained information, enhancing generalization. Additionally, a weight-based mechanism integrates these generators, ensuring compatibility with different types of anomalies	Encoder + Dilated CNN + GAN	Bench of NAB
	Song et al. [39]	2023	MEMTO is the first MVTs anomaly detection method using a Gated memory module for diverse normal pattern adaptation. Its two-phase training enhances stability and robustness across memory module-based models universally. Authors propose a bi-dimensional deviation-based criterion in online detection, aggregating info from latent and input space	Encoder and a weak form of Decoder of Transformer	SMD, MSL, SMAP, SWaT, PSM
	Kim et al. [40]	2023	The model includes a Transformer encoder, a 1D convolutional decoder, capturing multi-level information. Considering global trends and local variability, it predicts anomalies by comparing predicted and actual values, detecting anomalies above a specified threshold	Enc & Dec Transformer + 1D-CNN	Yahoo S5 and NeuIPS-TS benchmark
	Shin et al. [41]	2023	AnoFormer is a Transformer-based GAN with a generator and discriminator. It uses a two-step masking method to encode normal time-series data distribution, with entropy-based re-masking for uncertain parts	Encoder + GAN	NeuIPS, MIT-BIH Arrhythmia Database, 2D-Gesture and Power-Demand
Relative PE (Global attention)	Wang et al. [42]	2022	The authors proposed a variational Transformer-based model for unsupervised time-series anomaly detection, capturing correlations among features through an attention mechanism	Enc & Dec Transformer	SKAB and NAB

$$e_{ij} = \frac{(x_i W^Q)(x_j W^K)^T}{\sqrt{d_z}} \quad (1)$$

The projections $W^Q, W^K, W^V \in \mathbb{R}^{d_x \times d_z}$ are parameter matrices and are unique per layer [24]. where e_{ij} is an attention weight from positions j to i and is computed using a scaled dot-product.

$$\alpha_{i,j} = \frac{\exp(e_{ij})}{\sum_{k=1}^L \exp(e_{ik})} \quad (2)$$

Each coefficient weight $\alpha_{i,j}$ is calculated using the softmax function.

$$z_i = \sum_{j=1}^L \alpha_{i,j} (x_j W^V) \quad (3)$$

where z_i is the final computed output from the query, key, and value, denoted as multi-head attention, which performs self-attention computation in parallel in h the number of attention heads. Following this, outputs are linearly transformed and concatenated. The recursive model definition denotes z_i as the output of block i . As illustrated in Fig. 1, data processing progresses from the input layer through modules including attention, feedforward, and normalization, to the output layer. The feed-forward module converts self-attention outputs into final model outputs, functioning as a multi-layer perceptron. The Add & Norm module combines residual connections and layer normalization [44], ensuring robust gradient flow and computational balance. Here, sigmoid activation benefits the model for anomaly detection in training outputs, distinguishing between values 0 and 1.

4 Positional encoding (PE)

PE is crucial in designing sequential systems using attention mechanisms. In Transformers, PE compensates for the lack of inherent sequence order, distinguishing it from recurrent and Convolutional Neural Networks (CNN) [24]. The Encoder layer features self-attention and position-wise feedforward components, both permutation-equivariant and insensitive to input order. To give models positional awareness and keep to understand the order of input data, positions are typically embedded using PEs P , combined with input observations O as $(O + P)$ [45]. These embeddings encode specific positions of data points. Initially, Absolute PE used sinusoidal functions to preserve sequence order in NLP [24]. Recent advancements introduced Relative PE [28, 29] and Rotary PE [30] as superior alternatives in NLP. However, no comparative studies have assessed their performance in anomaly detection for time-series data.

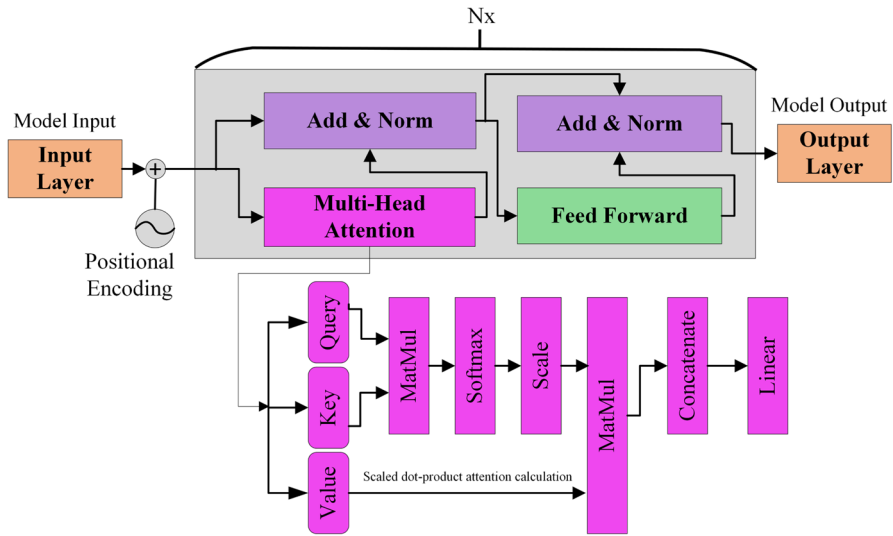


Fig. 1 Architecture of the encoder layer of the transformer

4.1 Absolute PE

The original Transformer architecture uses Absolute PE [24], which adds positional information $P = (p_1, \dots, p_L)$ to the input data x . The encoded output is $x_i = x_i + p_i$, computed in the input layer before the attention mechanism. Absolute PE assigns a unique fixed-length vector to each time-step in the input sequence, added to the tensor sequence before inputting into the Transformer. Equation 4 typically generates these vectors, encoding position information continuously and meaningfully. In the Eq. 4, $\sin$ is used to generate values for $2i$ or even time-step data; while, $\cos$ is used to generate values for $2i + 1$ or odd time-step observations.

$$\begin{aligned} \text{PE}_{(\text{pos}, 2i)} &= \sin\left(\frac{\text{pos}}{1000^{2i/d_{\text{model}}}}\right) \\ \text{PE}_{(\text{pos}, 2i+1)} &= \cos\left(\frac{\text{pos}}{1000^{2i/d_{\text{model}}}}\right) \end{aligned} \quad (4)$$

4.2 Relative PE

Relative PE, noted for its robustness, has been adapted in various forms. Shaw et al. [28] introduced a representation, later refined by Huang et al. [29] for long sequences, forming the basis of widely used architectures. Relative PE can be applied directly to Key and Query in the attention module, referred to as Representative attention [28] and Global attention [29]. Further details are provided in the following subsections.

4.2.1 Representative attention

The concept of Representative attention, proposed by Shaw et al. [28], forms the basis of Relative PE in NLP and computer vision tasks. Unlike Absolute PE, Representative attention extends self-attention to capture pairwise relationships among input elements. This mechanism models the input as a labeled, directed, fully connected graph, allowing relative distances between input elements x_i and x_j to be encoded into vectors $p_{(i,j)}^O, p_{(i,j)}^K, p_{(i,j)}^V \in \mathbb{R}^{d_z}$. These vectors are integrated into the self-attention module, modifying Eqs. 1–3 in the above section as described Eqs. 5–7:

$$e_{ij} = \frac{(x_i W^O + p_{(i,j)}^O)(x_j W^K + p_{(i,j)}^K)^T}{\sqrt{d_z}} \quad (5)$$

$$\alpha_{(i,j)} = \frac{\exp(e_{ij})}{\sum_{k=1}^L \exp(e_{ik})} \quad (6)$$

$$z_i = \sum_{j=1}^L \alpha_{(i,j)} (x_j W^V + p_{(i,j)}^V) \quad (7)$$

By doing so, the pairwise positional relation is trained during Transformer training. Relative information in Representative attention is supplied to the model on two levels: values and keys. Initially, relative information is included in the model as an additional component of the keys. The *softmax* operation as Eq. 2 remains unchanged from vanilla self-attention. Subsequently, relative information is reintroduced as a sub-component of the values matrix. Additionally, the authors suggest that Representative attention information becomes less useful beyond a certain distance, leading them to introduce a *clip* function to reduce the number of parameters. The encoding is formulated as follows to consider the distance between inputs i and j in computing their attention from Eq. 8 and 10, here the softmax operation remains unchanged as Eq. 2; while, *clip* function is computed as Eq. 11:

$$e_{ij} = \frac{(x_i W^O)(x_j W^K + p_{\text{clip}(i-j,k)}^K)^T}{\sqrt{d_z}} \quad (8)$$

$$\alpha_{(i,j)} = \frac{\exp(e_{ij})}{\sum_{k=1}^L \exp(e_{ik})} \quad (9)$$

$$z_i = \sum_{j=1}^L \alpha_{(i,j)} (x_j W^V + p_{\text{clip}(i-j,k)}^V) \quad (10)$$

$$\text{clip}(x, k) = \max(-k, \min(k, x)) \quad (11)$$

where p^V and p^K are the trainable weights of Representative attention on values and

keys, respectively. $p^V = (p_{-k}^V, \dots, p_k^V)$ and $p^K = (p_{-k}^K, \dots, p_k^K)$ where $p_i^V, p_i^K \in \mathbb{R}^{d_z}$. The scaler k is the maximum relative distance.

4.2.2 Global attention

The Representative attention, introduced by Shaw et al. [28], was deemed impractical for long sequences due to the additional relative position encoding it required, leading to high time complexity and memory usage. To overcome these issues, Haung et al. [29] refined Representative attention, which is a more robust mechanism called Global attention. In Eq. 5, above section we can observe that the Representative attention requires $O(L^2d)$ memory usage. In Global attention, a new method is introduced, termed vector-based, for computing Relative PE. This method reduces the intermediate memory requirement from $O(L^2d)$ to $O(Ld)$ using *skewing* operations [29]. According to this paper, the authors also omitted the additional Relative PE corresponding to the value term and focused solely only on the key component. The encoding is formulated as Eq. 12, the softmax is remind unchanged as Eq. 2, the skewing is computed as Eq. 13:

$$e_{ij} = \frac{(x_i W^Q)(x_j W^K)^T + S^{\text{rel}}}{\sqrt{d_z}} \quad (12)$$

$$S^{\text{rel}} = \text{Skew}(W^Q P) \quad (13)$$

where S^{rel} is the skew-symmetric matrix generated by W^Q and P .

4.3 Rotary PE

Rotary PE uses rotation matrices to encode absolute positions and incorporates explicit relative position dependencies in self-attention mechanisms [30]. This method applies rotations to query and key representations based on their absolute positions before computing dot-product attention, ensuring that attention calculations depend only on relative distances between data points. The use of complex exponential functions introduces rotational transformations to PEs, enhancing the model's ability to capture detailed sequence positions. Rotary PE combines the benefits of both Absolute PE and Relative PE. The rotation of key and query matrices is modified the scaled dot-product attention calculation from Eqs. 1–3 to Eqs. 14–16.

$$e_{ij} = \frac{(x_i W^Q + R_n \theta)(x_j W^K + R_n \theta)^T}{\sqrt{d_z}} \quad (14)$$

$$\alpha_{(i,j)} = \frac{\exp(e_{ij})}{\sum_{k=1}^L \exp(e_{ik})} \quad (15)$$

$$z_i = \sum_{j=1}^L \alpha_{(i,j)} (x_j W^V) \quad (16)$$

Here, $R_n\theta$ is a rotation matrix that rotates x by $n\theta$ radians:

$$R_n\theta = \begin{bmatrix} \cos(n\theta) & -\sin(n\theta) \\ \sin(n\theta) & \cos(n\theta) \end{bmatrix}$$

where n is the position of the data, and θ is a scalar value (angle) chosen to rotate the value.

5 Experimental evaluation

5.1 Dataset

DL-based anomaly detection models often require large data volumes, posing a challenge in finding suitable experimental data. We tested the effectiveness of PEs on a public dataset from an Industrial Control System (ICS) augmented with a Hardware-In-the-Loop (HIL) simulator, emulating steam turbine and pumped-storage hydropower generation [46]. The HIL-based Augmented ICS (HAI) data is an MVTAD dataset and has been available since February 2020. We used version 23.05, collected from 2022-08-17 to 2022-08-19 sly, containing 230,400 data points and 226 features, with labels indicating normal (0) or abnormal (1) states. The original dataset, highly imbalanced with a 96:4 normal-to-abnormal ratio, affects model performance. To balance it, we applied random under-sampling, adjusting the ratio to 55:45 and reducing the dataset to 18,906 data points. This balanced dataset allows the model to effectively learn the properties of both normal and abnormal data.

5.2 Data preprocessing and feature engineering

The generalization performance of supervised anomaly detection models often depends on effective data preprocessing. This includes noise elimination, handling missing data, and converting symbolic features into numerical ones readable by ML models, which is crucial in practice. In this study, we ensured robust performance estimations of proposed PEs on the Encoder Transformer by performing basic preprocessing steps, feature engineering, and scaling data between 0 and 1. After removing null and missing values, our dataset contained 226 high-dimensional features, necessitating feature engineering to select a valuable subset. We applied the VarianceThreshold (VT) method, which identified 151 features with constant values (0 and 1), leaving 75 features. We then used the Pearson Correlation Coefficient (PCC) to remove 37 highly correlated features (correlation 90%). Finally, we applied Information Gain (IG) to measure mutual dependence between variables, identifying and removing 2 features with IG values of zero. As shown in Fig. 2, our dataset default contained 226 features. After the first feature selection step with VT, 75

features remained. Following the second step with PCC, 38 features remained. After the third step with IG, we obtained the 36 most useful features for model evaluation.

As another aspect, the division of the entire available dataset into appropriate ratios is a critical step for effective model training and generalization. Therefore, a ratio of 70% for the training set, 15% for the validation set, and 15% for the test set was chosen, significantly influencing the model's performance accuracy on the test set.

5.3 Evaluation metrics

In DL model performance evaluation, metrics quantify how well the model performs on a given task. This study focuses on point anomaly detection for multivariate time-series data, treating it as a supervised anomaly detection problem similar to binary classification. In anomaly detection, evaluation metrics typically rely on the confusion matrix. It includes True Positive (TP), False Positive (FP), False Negative (FN), and True Negative (TN). In this study, we propose six commonly used metrics to evaluate the model's PEs: accuracy, precision, recall, F1-score, AUC-ROC, and Cohen's Kappa. These metrics reflect how predictions align with labels.

Accuracy shows the proportion of correctly classified instances among the total number of instances. It is a general indicator of the model's ability to correctly identify both normal and abnormal instances. However, it might not be a reliable metric in imbalanced datasets, as in anomaly detection.

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \quad (17)$$

Precision shows the proportion of correctly predicted anomalies (TP) out of all predicted anomalies (TP + FP). It measures how accurate the positive predictions

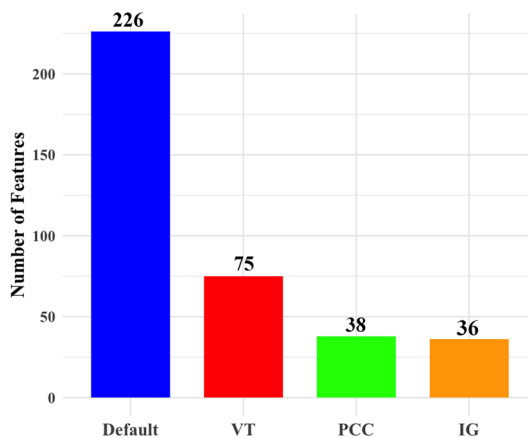


Fig. 2 Feature selection steps reducing the features from 226 to 36

are. In anomaly detection, a higher precision indicates fewer FP (normal instances mistakenly identified as abnormal).

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (18)$$

Recall shows the proportion of correctly predicted anomalies (TP) out of all actual anomalies (TP + FN). It measures the model's ability to identify all true anomalies in the dataset. In anomaly detection, a higher recall indicates that the model successfully detects most of the actual anomalies, resulting in fewer missed anomalies (FN).

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (19)$$

F1-score shows the harmonic mean of Precision and Recall. The F1-Score provides a single metric that balances Precision and Recall. It is useful when there is an uneven class distribution and is a good indicator when both FP and FN are costly.

$$\text{F1-score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (20)$$

AUC (Area Under the Curve) measures the area under the Receiver Operating Characteristic (ROC) curve, which plots the TP rate (Recall) against the FP rate. An AUC of 1 indicates a perfect model; while, an AUC of 0.5 indicates a random classifier. A higher AUC means the model is better at distinguishing between normal and abnormal classes.

Cohen's Kappa measures the agreement between two raters (in this case, the predicted classes and the actual classes), adjusting for the chance of random agreement. Kappa ranges from -1 to 1, where: 1 indicates perfect agreement and 0 indicates agreement purely by chance. Negative values indicate worse-than-random agreement. Kappa is used in this study for anomaly detection to assess the significant imbalance between normal and abnormal instances.

$$\text{Kappa} = \frac{\text{Observed Agreement} - \text{Expected Agreement}}{1 - \text{Expected Agreement}} \quad (21)$$

5.4 Hyperparameter optimization

Hyperparameter tuning is crucial for training ML algorithms, particularly in DL due to their complex architectures. Optuna is highly efficient for this purpose, employing Bayesian optimization and genetic algorithms, as highlighted in a recent comparative analysis of state-of-the-art optimization techniques in [47]. It dynamically adjusts parameters. In this experiment, we conducted 100 trials over 10 epochs, during which Optuna determined the optimal values in Trial 84 from the provided ranges, as detailed in Table 2: 2 heads (from a range of 1–8), 200 dimensions (from 200 to 512), 8 layers (from 1 to 10), a maximum length of 3004 (from 1000 to 5000), a feedforward size of 1981 (from 16 to 2048), a dropout rate of 0.00025 (from 0.0–0.5), a learning rate of $3.70\text{e}-05$ (from $1\text{e}-5$ to $1\text{e}-2$), and a batch size

Table 2 Optimized parameter values

Parameters	Ranges	Optimal values
# head	1–8	2
d_model	200–512	200
# layer	1–10	8
Max_len	1000–5000	3004
ff_hidden	16–2048	1981
drop_rate	0.0–0.5	0.00025
lr_rate	1e–5–1e–2	3.70e–05

manually set to 32 for the entire sequence length experiment to ensure effective learning and high accuracy.

6 Experimental result

To better understand the efficiency of PEs, we evaluated the model from two perspectives: classification and training time scores. Therefore, this section has two subsections. The first subsection discusses the metric scores, and the second provides details of the PEs for the required training time per epoch. It helps decision-makers consider both factors when developing attention-based mechanisms.

6.1 Classification result of PEs

Sequences refer to the window size; in time-series data collected daily, hourly, or weekly, the window size must align with the collected trends. For instance, if our dataset is collected hourly and we aim to predict anomalies and normal points one day ahead, we must set the sequence length to 24 h. The sequence length is crucial for each PE because the length of coding in time-series data depends on the sequence length we want to process through models. Therefore, in this study, we experimented with the models' performance using sequence lengths of 8, 16, 32, 64, and 128 to address researchers' questions regarding which PE is better for different lengths of sequences.

Table 3 compares the performance of four PE models (Absolute PE, Rotary PE, Representative attention, and Global attention) on MVTAD problems across different sequence lengths, considering 8, 16, and 32 as short sequences and 64 and 128 as long sequences, using various metrics (accuracy, precision, recall, F1-score, AUC, and Kappa). Absolute PE demonstrates consistently high performance across all metrics for short and long sequences, particularly high in accuracy, precision, recall, and AUC. Its accuracy peaks at a window size of 8 (0.985) and is lowest at 128 (0.981). Precision is highest at 8 (0.980) and lowest at 128 (0.972). Recall remains consistently high, peaking at 16 (0.989) and slightly lower at 128 (0.988). The F1-score is also stable, with the highest at 8 (0.984) and the lowest at 128 (0.980). The AUC follows a similar pattern, peaking at 16 (0.986) and lowest at 128 (0.982).

Table 3 Performance comparison of proposed PEs for different sequence lengths

PE	Metrics	Sequences (window size)				
		8	16	32	64	128
Absolute	Accuracy	0.985	0.985	0.983	0.984	0.981
	Precision	0.980	0.977	0.977	0.979	0.972
	Recall	0.988	0.989	0.988	0.988	0.988
	F1-score	0.984	0.983	0.983	0.984	0.980
	AUC	0.985	0.986	0.984	0.985	0.982
	Kappa	0.979	0.969	0.976	0.979	0.973
Rotary	Accuracy	0.973	0.978	0.967	0.976	0.971
	Precision	0.958	0.968	0.945	0.962	0.954
	Recall	0.985	0.984	0.983	0.985	0.982
	F1-score	0.971	0.976	0.964	0.974	0.968
	AUC	0.975	0.979	0.969	0.977	0.972
	Kappa	0.956	0.965	0.943	0.961	0.950
Representative attention	Accuracy	0.983	0.982	0.978	0.978	0.971
	Precision	0.982	0.979	0.969	0.969	0.953
	Recall	0.983	0.983	0.984	0.982	0.984
	F1-score	0.982	0.981	0.987	0.976	0.969
	AUC	0.983	0.982	0.979	0.978	0.973
	Kappa	0.986	0.974	0.966	0.965	0.952
Global attention	Accuracy	0.962	0.967	0.971	0.972	0.973
	Precision	0.939	0.954	0.968	0.958	0.959
	Recall	0.979	0.973	0.969	0.981	0.983
	F1-score	0.959	0.963	0.969	0.969	0.971
	AUC	0.964	0.967	0.971	0.972	0.974
	Kappa	0.934	0.943	0.952	0.953	0.956

Kappa is highest at 8 (0.979) and lowest at 128 (0.973), indicating that Absolute PE performs better for short sequences than long sequences.

Rotary PE, on the other hand, shows lower performance with the highest accuracy at 16 (0.978) and the lowest at 32 (0.967). Precision fluctuates, peaking at 16 (0.968) and dropping at 32 (0.945). Recall is stable and high across window sizes, peaking at 64 (0.985). The F1-score peaks at 16 (0.976) but is lowest at 32 (0.964). AUC is highest at 16 (0.979) and lowest at 32 (0.969). Kappa follows a similar trend, peaking at 16 (0.965) and lowest at 32 (0.943), indicating it performs better on medium sequence lengths but is not as outstanding as Absolute PE.

Representative Attention shows high performance with accuracy slightly decreasing as the window size increases, highest at 8 (0.983) and lowest at 128 (0.971). Precision is stable and highest at 8 (0.982), lowest at 128 (0.953). Recall remains high across all window sizes, peaking at 128 (0.984). The F1-score is highest

at 32 (0.987) and lowest at 128 (0.969). AUC is highest at 8 (0.983) and lowest at 128 (0.973). Kappa is highest at 8 (0.986) and lowest at 128 (0.952), indicating that Representative attention is better for short sequences compared to long sequences.

Global attention shows a trend of increasing performance with larger window sizes. Accuracy increases steadily, peaking at 128 (0.973). Precision is highest at 32 (0.968) and lowest at 8 (0.939). Recall remains high, peaking at 128 (0.983). The F1-score follows a similar trend, highest at 128 (0.971). AUC is highest at 128 (0.974), showing steady improvement with increasing window size. Kappa peaks at 128 (0.956) and is lowest at 8 (0.934).

In the case of comparing with each other Absolute PE is the best overall model for MVTAD, given its high and consistent scores across various metrics and window sizes. Representative attention also performs very well, especially in recall and F1-score, with high stability across window sizes. Global attention performs better with larger window sizes, indicating robustness in such scenarios. It allows the model to consider all elements in a sequence at once, which is especially powerful for long sequences where understanding relationships across a wide range is important. In this context, it effectively captures long-range dependencies, making it valuable when context from distant parts of the sequence matters. For shorter sequences, relationships are simpler; therefore, Absolute PE and Representative attention can provide sufficient positional information without the need to handle all inputs at once. This is beneficial when distant dependencies are less significant, as shorter sequences have limited context, reducing the risk of information overload. Absolute PE and Representative attention can encode positions and relationships effectively without adding the complexity seen in Global attention, which may offer little value for shorter sequences. Rotary PE, while decent in performance, is generally outperformed by the other models in most metrics.

To clarify the behavior of PEs, Fig. 3 illustrates the confusion matrix for (a) Absolute PE, (b) Rotary PE, (c) Representative attention, and (d) Global attention, respectively. The confusion matrix for the anomaly detection model using Absolute PE, shown in Fig. 3(a), indicates strong performance. The model correctly identified 1205 abnormal instances (TP) and 1554 normal instances (TN). There were only 20 normal instances misclassified as abnormal (FP) and 5 abnormal instances misclassified as normal (FN). This high number of TP and TN suggests that the model is highly reliable in distinguishing between normal and abnormal data points. The low counts of FP and FN indicate minimal mistakes, crucial for practical applications where misclassification can have significant consequences. Similarly, the confusion matrix for the anomaly detection model using Rotary PE in a Transformer, shown in Fig. 3(b), also demonstrates strong performance. The model correctly identified 1203 abnormal instances (TP) and 1544 normal instances (TN). There were 30 normal instances misclassified as abnormal (FP) and 7 abnormal instances misclassified as normal (FN). Although the number of FP (30) and FN (7) is slightly higher compared to Absolute PE, it remains relatively low, indicating the model's reliability in distinguishing between normal and abnormal data points. The confusion matrix for the anomaly detection model using Representative attention, shown in Fig. 3(c), also highlights strong performance. The model correctly identified 1201

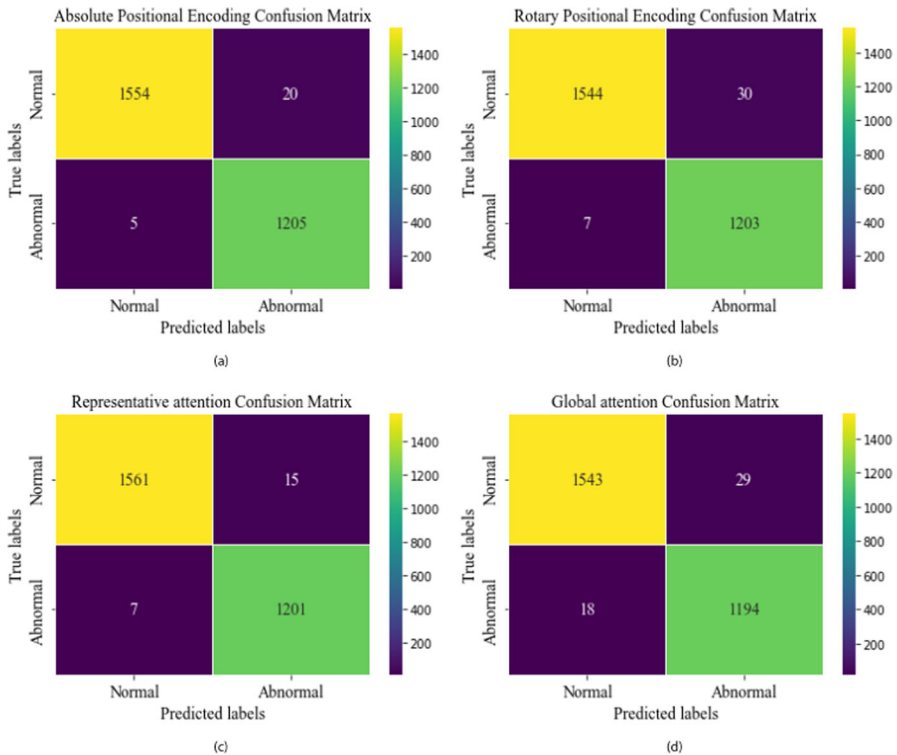


Fig. 3 Confusion matrix for **a** Absolute PE, **b** Rotary PE, **c** Representative attention, and **d** Global attention

abnormal instances (TP) and 1561 normal instances (TN). There were 15 normal instances misclassified as abnormal (FP) and 7 abnormal instances misclassified as normal (FN). The high number of TP's and TN's underscores the model's reliability; while, the low number of FP (15) and FN (7) suggests precision and effectiveness in detecting anomalies. Finally, the confusion matrix for the Global attention model in MVTAD, shown in Fig. 3(d), demonstrates excellent performance. The model accurately identifies 1543 normal instances and 1194 abnormal instances, with minimal errors. Only 29 normal instances were misclassified as abnormal and 18 abnormal instances were misclassified as normal.

The comparative analysis of the different PEs demonstrates their varying effectiveness in anomaly detection based on key performance metrics. Absolute PE showed high reliability, with an accuracy of 99.1%, precision of 98.4%, recall of 99.6%, and an F1-score of 0.990. This indicates its strong capability to accurately distinguish between normal and abnormal instances with minimal misclassification. Rotary PE, while still effective, showed a slightly lower accuracy of 98.7%, precision of 97.6%, recall of 99.4%, and an F1-score of 0.985, indicating a minor increase in FP and FN compared to Absolute PE. The model using Representative attention performed similarly to Absolute PE, achieving the highest accuracy of 99.2%, a

precision of 98.7%, and recall of 99.4%, with an F1-score of 0.990, highlighting its balanced performance in anomaly detection. On the other hand, the Global attention model achieved an accuracy of 98.3% and a recall of 98.5%, but its precision was slightly lower at 97.6%, resulting in an F1-score of 0.980. This suggests a slight increase in misclassified instances compared to the other models. Representative attention and Absolute PE demonstrated the most balanced and robust performance, making them highly effective for anomaly detection tasks. Rotary PE and Global Attention, while still reliable, have slightly higher error rates, impacting their overall performance. This comparative analysis quantitatively clarifies the effectiveness of each PE method, facilitating in the selection of the most appropriate mechanism based on specific requirements for accuracy, precision, and recall.

6.2 Classification result of DL models

To demonstrate the effectiveness of Transformer models and provide strong evidence supporting the clarity of our study, we evaluated the performance of non-self-attention DL models on the HAI dataset for the MVTAD problem. Table 4 shows the performance results of the current state-of-the-art DL models, including 1D-CNN, RNN, LSTM, and GRU, used for the MVTAD problem.

As seen in Table 4, GRU model consistently delivers the best performance across all sequence lengths. GRU achieves the highest values for metrics such as accuracy, precision, recall, F1-score, AUC, and Kappa score, highlighting its superior effectiveness compared to the other models. Notably, its accuracy remains high, starting at 0.967 for a sequence length of 8 and only slightly decreasing to 0.913 for a sequence length of 128, demonstrating its stability across varying sequence lengths. LSTM also performs well, especially for shorter sequence lengths like 8, 16, and 32, where it maintains strong accuracy, precision, and F1-score values. However, its performance declines when dealing with longer sequences (64 and 128), although it still outperforms both RNN and 1D-CNN. RNN model produces mixed results, with relatively stable accuracy values at shorter sequence lengths but significantly lower recall and F1-scores when compared to GRU and LSTM, indicating its limited effectiveness in distinguishing abnormal points from normal ones. 1D-CNN model displays the lowest performance across all metrics in this comparison. Its accuracy gradually decreases from 0.819 for a sequence length of 8 to 0.704 for a sequence length of 128, with its precision and F1-score consistently declining as the sequence length increases. This suggests that 1D-CNN struggles more with longer sequences, making it the least suitable model for anomaly detection tasks than others.

While all models show a trend of decreasing performance with longer sequence lengths, GRU emerges as the most reliable and effective model for classifying normal behaviors from abnormal ones in multivariate time-series data, followed by LSTM. RNN and 1D-CNN models perform relatively worse, with 1D-CNN being the least effective. Given these results, GRU is recommended for tasks involving similar data and anomaly detection needs.

In comparing non-self-attention DL with Transformers, we can observe that, across all sequence lengths, attention-based models exhibit superior performance. According to Tables 3 and 4, we do not recommend RNN and 1D-CNN for the

Table 4 Performance comparison of DL models for different sequence lengths

Models	Metrics	Sequences (window size)				
		8	16	32	64	128
1D-CNN	Accuracy	0.819	0.770	0.756	0.731	0.704
	Precision	0.806	0.783	0.747	0.721	0.695
	Recall	0.720	0.684	0.630	0.605	0.610
	F1-score	0.760	0.730	0.683	0.658	0.649
	AUC	0.850	0.749	0.741	0.732	0.710
	Kappa	0.587	0.506	0.492	0.481	0.477
RNN	Accuracy	0.818	0.820	0.822	0.787	0.739
	Precision	0.824	0.845	0.824	0.817	0.777
	Recall	0.668	0.652	0.641	0.615	0.605
	F1-score	0.738	0.736	0.721	0.702	0.680
	AUC	0.801	0.802	0.798	0.770	0.760
	Kappa	0.522	0.520	0.519	0.501	0.429
LSTM	Accuracy	0.859	0.840	0.836	0.821	0.814
	Precision	0.896	0.863	0.827	0.817	0.806
	Recall	0.784	0.734	0.730	0.715	0.716
	F1-score	0.836	0.793	0.775	0.763	0.758
	AUC	0.841	0.831	0.830	0.827	0.807
	Kappa	0.596	0.581	0.580	0.504	0.507
GRU	Accuracy	0.967	0.960	0.950	0.919	0.913
	Precision	0.927	0.918	0.923	0.914	0.912
	Recall	0.944	0.863	0.740	0.860	0.814
	F1-score	0.935	0.890	0.821	0.886	0.860
	AUC	0.964	0.959	0.925	0.910	0.917
	Kappa	0.727	0.727	0.710	0.722	0.716

MVTAD problems. Additionally, Table 4 shows that all DL models perform better with shorter sequences than with longer ones. As the window size increases, the metric scores decrease, indicating that the models struggle to distinguish between normal and abnormal points. However, in the case of Transformers, where we evaluated the efficacy of different PEs, clear trends emerge, suggesting that implementing certain PEs during the architecture design of Transformers can improve performance for short sequences; while, other PEs are more suitable for longer sequences.

6.3 Training time result

Estimating the time consumption of attention-based models is crucial for decision-makers to consider during model development and training. Additionally, an interesting belief persists among industries employing attention-based models for time-series anomaly detection: that their training times exceed those of RNN-based

neural networks. To clarify this misconception, we assessed the training times of PEs as a subset of performance evaluation across short and long sequences, using the tqdm python library to track progress during each epoch of training. Based on the provided training time data for each model per epoch across different sequence lengths (window sizes) in Table 5: Absolute PE shows relatively consistent training times across all sequence lengths, ranging from 25 s for the shortest sequence (8) to 192 s for the longest sequences (64 and 128). This consistency suggests that Absolute PE maintains a stable training performance regardless of the sequence length, with a gradual increase in time as the sequence length grows. Rotary PE demonstrates slightly longer training times compared to Absolute PE, with times ranging from 29 s for the shortest sequence (8) to 242 s for the longest sequence (128). The training times for Rotary PE generally increase with sequence length, indicating that it may require more computational resources as the input sequence grows. Representative attention starts with 48 s for the shortest sequence (8) and scales up to 366 s for the longest sequence (128). This substantial increase in training time suggests that Representative attention is more computationally intensive, particularly noticeable in longer sequences, where it requires significantly more time compared to other models. Global attention exhibits training times ranging from 33 s for the shortest sequence (8) to 212 s for the longest sequence (128). The training times for Global attention are generally lower across different sequence lengths, indicating efficient training performance across a range of inputs. In the case of comparing with each other, Absolute PE and Rotary PE generally demonstrate more efficient training times compared to Representative attention, which exhibits the longest training times, especially for longer sequences. Global attention achieves a balance, showing low training times across different sequence lengths. Efficient training times are crucial for scalability and real-world deployment, where models like Absolute PE and Global attention may offer advantages in scenarios requiring faster training and inference times.

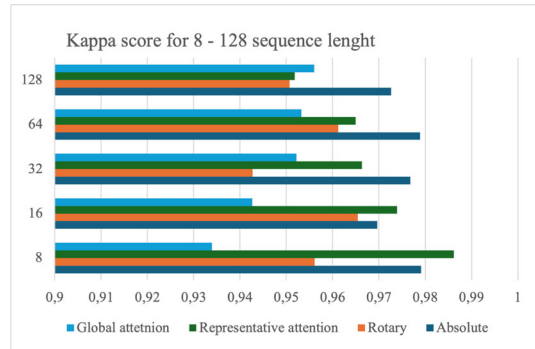
6.4 Discussion

In Table 3 of the results, the Kappa scores highlight clear trends for the current mechanism. To illustrate this, Fig. 4 compares the Kappa scores of four proposed PE methods Absolute PE, Rotary PE, Representative Attention, and Global attention-

Table 5 Training time (seconds) for different sequences (window sizes)

PE	Sequences (window size)				
	8	16	32	64	128
Absolute	25	53	69	126	192
Rotary	29	55	75	107	242
Representative attention	48	55	77	185	366
Global attention	33	37	57	100	212

Fig. 4 Comparison of Kappa scores for sequence lengths ranging from 8 to 128



used in Transformer neural networks across varying sequence lengths (8, 16, 32, 64, and 128). For short sequences (8 and 16), Representative attention consistently exhibits the highest Kappa scores, suggesting its superior performance in handling short input sequences. Absolute PE also performs well for short sequences but is slightly lower than Representative attention. Rotary PE, while showing low Kappa scores, does not perform as well as the others for short sequences. Global attention has the lowest Kappa scores for short sequences, indicating it is less effective in such cases. When examining medium sequence lengths (32), Absolute PE maintains high Kappa scores, slightly better than its performance on short sequences. Rotary PE and Representative attention remain consistent with their short sequence performance, with Rotary PE being low and Representative attention high. Global attention shows improvement at this sequence length, achieving a Kappa score similar to that of the other methods. For long sequences (64 and 128), Absolute PE achieves its highest Kappa scores, indicating its robustness and reliability for longer sequences. Rotary PE also improves with longer sequences but remains slightly below Absolute PE. Representative attention's performance remains high but does not increase as significantly with longer sequences as Absolute PE. Interestingly, Global attention, which had the lowest scores for short sequences, performs exceptionally well with long sequences, matching or even surpassing Absolute PE at sequence lengths of 64 and 128. In summary, we can observe from analyzing that Representative attention is best suited for short sequences due to its consistently high Kappa scores; while, Absolute PE is the most effective for long sequences, demonstrating superior performance across all sequence lengths, especially longer ones. Global attention,

Table 6 Comparison of PEs

PE	Parameter size	Memory	Complexity
Absolute PE	Ld_{model}	Ld_{model}	Ld_{model}
Rotary PE	Ld_{model}	L^2d_z	L^2d_z
Representative attention	$(2L - 1)d_z$	$L^2d_z + L^2$	L^2d_z
Global attention	Ld_z	$Ld_z + L^2$	L^2d_z

although less effective for short sequences, performs exceptionally well with longer sequences, making it a strong contender for handling long input sequences in Transformer neural networks.

Typically, the performance and training time of PEs depends on parameter size, space complexity, and time complexity. Therefore, Table 6 compares proposed PE methods based on these parameters to assist decision-makers. Absolute PE scales linearly with both sequence length and model dimension (Ld_{model}), making it efficient and manageable across varying sequence lengths. Rotary PE has a linear parameter size (Ld_{model}) but quadratic memory and computational complexity (L^2d_z), requiring more resources for longer sequences. Representative attention, with a parameter size of $(2L - 1)d_z$, has memory usage and computational complexity of $L^2d_z + L^2$ and L^2d_z , respectively, indicating high resource demands similar to Rotary PE. Global attention features a linear parameter size (Ld_z) but combines linear and quadratic terms for memory ($Ld_z + L^2$) and has a quadratic computational complexity (L^2d_z). For short sequences (8, 16), Absolute PE is the most efficient due to its linear scalability. Global attention also performs well for short sequences. For medium to long sequences (32, 64, 128), the choice among Global attention, Representative attention, and Rotary PE depends on trade-offs between memory usage, computational complexity, and performance benefits. Absolute PE is optimal for short sequences, while the choice among other methods for longer sequences depends on specific trade-offs.

7 Conclusion and future work

In this study, we examined the performance of Absolute PE, Rotary PE, as well as from Relative PE two commonly used modifications (Representative attention and Global attention) on the Encoder only Transformer networks for IoT-based MVTAD problems. Considering that the efficacy of PEs usually depends directly on the length of the sequence, we evaluated accuracy on sequences of lengths 8, 16, 32, 64, and 128, categorized as short and long sequences. For the experiments, we utilized the HAI security dataset. Hyperparameters are important for attention-based neural networks, so we conducted Optuna hyperparameter tuning, a lightweight algorithm for hyperparameter optimization. Our dataset with 226 features had many constants, so we conducted a three-step feature selection process including VT, PCC, and IG. As a result, the dimension of our dataset decreased from 226 to 36 valuable features. We evaluated the efficacy of PEs from two perspectives: classification performance and calculation of training time. For classification performance, we calculated the metrics values of the test set. To assess the training time, we utilized a Python built-in library which is called `tqdm` and include the training loop inside and run the training process to know which PEs how much time require on per epoch. To compare the performance of Absolute PE and Rotary PE, Absolute PE outperforms in short sequences; while, Rotary PE is efficient in long sequences. Regarding Relative PE modifications, Representative attention surpasses in short sequences; whereas, Global attention demonstrates good performance in long sequences. In conclusion, the comparative analysis shows that Absolute PE and Representative attention offer

the most balanced and reliable performance for anomaly detection, with high accuracy and low misclassification rates. While Rotary PE and Global attention are also effective, they have slightly higher error rates, making Absolute PE and Representative Attention preferable for tasks requiring greater precision and consistency. From a training time perspective, training times vary depending on the PEs utilized and the sequence length. Absolute PE exhibits longer training times as sequence length increases, peaking at 128 s. In contrast, Rotary PE generally requires less time, particularly for shorter sequences, though longer sequences, notably at 128 s, experience a rise in training time. Representative attention demands more time, especially for longer sequences; while, Global attention maintains low times across lengths, showing stable performance. Additionally, each PE has been analyzed comparatively in terms of parameter size, space complexity, and time complexity. When sorted from worst to best, considering these behaviors, Global attention ranks lowest, followed by Representative attention, Rotary PE, and Absolute PE, respectively, indicating that Absolute PE performs the best; whereas, Global attention ranks the lowest. The selection of PEs varies by problem. Our results show that in the architecture design of attention-based models, PE is crucial. For MVTAD context, (i) Absolute PE and Representative attention are best for short sequences, (ii) Rotary PE and Global attention are ideal for longer sequences, (iii) Absolute PE is optimal for fast inference with minimal training time, and (iv) Absolute PE and Rotary PE are excellent for minimizing time and space complexity.

In the point anomaly problem, we encounter two types of detection: binary classification and change point detection. While we evaluate PE performance within binary classification, change point detection typically relies on construction-based methods, where sequence length (or window size) plays a critical role. In this scenario, performance evaluation suggests further investigation. Additionally, evaluating the effectiveness of these PEs can be conducted in prediction and classification within the context of IoT-based multivariate time-series data in the future.

Author contributions Methodology and writing—reviewing contributed by Abdul Amir Alioghli and Feyza Yıldırım Okay; editing and supervision contributed by Feyza Yıldırım Okay; conceptualization, software and visualization contributed by Abdul Amir Alioghli. All authors have read and legally accepted the final version of the article published in the journal.

Data availability No datasets were generated or analyzed during the current study.

Declarations

Conflict of interest The authors declare no Conflict of interest.

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